

a.

Batch-Size: 3

Results: Lowest loss occurred at: 4.00, 0.21, -0.11, 4.00, 0.21, 4.00, 0.21

With standard deviation of 1.92

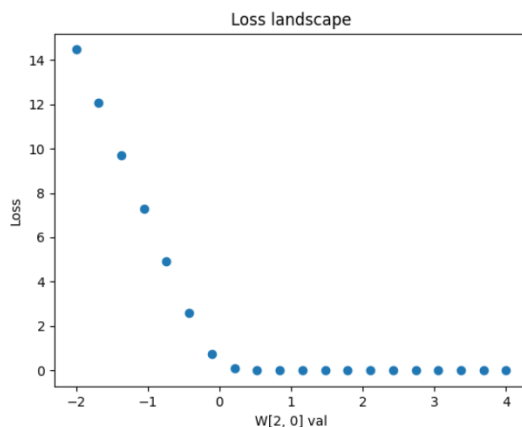
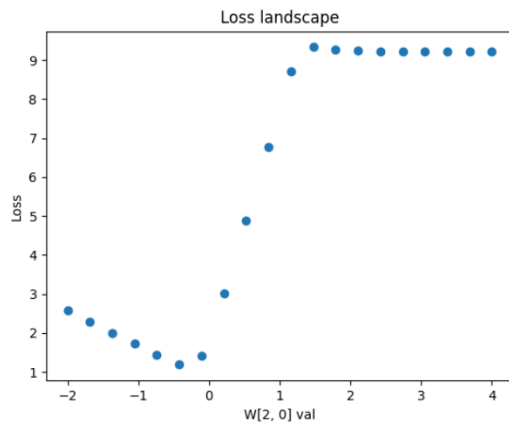
Batch-Size: 2

Results: Lowest loss occurred at: 4.00, 4.00, 0.21, -0.42, 2.11, 4.00, 0.21, 0.53, 4.00, 0.21

With standard deviation of 1.83

As the batch size increases, the standard deviation decreases. However, for some batch sizes, the standard deviation increases as seen in the progression results for batches 2 and 3. For smaller batch sizes, the loss landscape is more noisy and the optimization process is less stable. On the other hand, for larger batch sizes, the loss landscape is less noisy and the optimization process is more stable.

In the graphs below, we can see that there are significant jumps in the loss landscape for batch size 3 than batch size 2.



b.

When training a machine learning model using gradient descent, the choice of batch size is important. Smaller batch sizes converge faster but have more variance in the loss landscape, while larger batch sizes produce a smoother loss landscape but converge slower. Stochastic gradient descent is the fastest method but can be unstable due to high variance in gradients. Mini-batch gradient descent offers a compromise between stochastic and batch gradient descent, providing smooth convergence and faster convergence than batch gradient descent. Batch gradient descent is the most stable but slowest method. The choice of method and batch size depends on the problem and trade-off between convergence speed, stability, and computational efficiency.